

Rudrajit Choudhuri

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Summary

Mixed-methods UX and HCI researcher with 6+ years of experience leading end-to-end research on Human-Centered AI, Software Engineering, Cognitive Science, UX Design, Future of Work, and CS Education. Two Microsoft research internships (Microsoft Research, SAINTES; Microsoft Tech Futures, X3) delivering large-scale studies with 2,000+ participants, design frameworks, and product guidance adopted by partner teams. 33 publications at leading SE/HCI venues (ICSE, TOSEM, ICSME, ICER, EMSE, and ACMQ) and machine learning and computer vision venues (CVIP, ICVGIP, Applied Soft Computing, and Neural Computing and Applications), with four distinguished paper awards. Research covered by media and featured in industry newsletters. Currently focused on (1) cognitive and socio-technical factors shaping human-AI collaboration, and (2) designing human-centered interfaces for AI tools that improve AI-native knowledge work.

Education

Oregon State University

Corvallis, OR

Ph.D. in Computer Science (in progress), GPA 3.9/4.0. Focus: Human-Centered AI, Software Engineering

M.S. in Computer Science, GPA 4.0/4.0. Focus: Human-Centered AI, CS Education

Committee: A. Sarma (Advisor), M. Burnett, C. Hundhausen, I. Steinmacher, C. Sanchez

St. Thomas' College of Engineering & Technology

Kolkata, India

B.Tech (Hons.) in Computer Science, GPA 9.81/10, Rank 1 in college

Advisor: A. Halder. Focus: Machine Learning & Deep Learning in Radiology, Biomedicine, and Robotics

Experience

Graduate Research Assistant

Sep 2022 – Present

Oregon State University, Advisor: Anita Sarma

Corvallis, OR

Except Summer 2024; Spring 2025 (fellowship); Summers 2025 and 2026 (Microsoft internships)

- Led user research on generative AI's effects on productivity, cognition, and self-efficacy; mapped AI failure modes and their causes to concrete design strategies for safer, more effective human-AI interaction (ICSE 2024).
- Built an empirically grounded model of the factors shaping developers' trust in and adoption of AI, through a large-scale mixed-methods study with developers at GitHub and Microsoft (N=238); published at ICSE 2025 and extended in ACM TOSEM.
- Modeled how routine AI reliance leads to cognitive debt among STEM students across five universities (N=300), producing interface and curriculum interventions that preserve reflection and critical thinking; covered by TV and news media.
- Led user research with CS students on where and why they struggle to use AI in learning, detailing impacts on learning, task quality, self-perception, and adoption decisions (CSEE&T 2025).
- Co-developed a framework modeling how interpersonal challenges affect the sense of welcomeness in open source, through a study of 706 Linux Foundation contributors, producing actionable practices for inclusive communities (ICSE 2025).

Applied Science Intern, Tech Futures X3

Jun 2026 – Aug 2026

Microsoft, Mentors: Alice Ferng, Cinoo Lee, Peter Jahn

Redmond, WA

- Led a large-scale investigation into what makes a great co-worker in an AI-native workplace (22 interviews; mixed-methods survey of 1,534 knowledge workers). Built the BACI framework of co-worker qualities that apply to both humans and AI, identified 11 co-worker archetypes and how priorities vary across individuals, and a taxonomy of AI work etiquette for sharing AI-supported work.
- Transformed years of human-centered AI research into the [Design Cookbook for UI/UX of AI](#), an interactive resource of 80 design patterns across 10 design dimensions mapped to 21 human-factors challenges, bridging research and product design.
- Led an empirical study with 448 developers on where and why they draw the line on AI autonomy; showed how task identity, accountability, and cognitive demands shape delegation decisions, and derived implications for agentic tooling and meaningful work design.
- Co-led an investigation into agent-to-agent coordination in large-scale multi-agent workplaces.

PhD Research Intern, SAINTES Group

Jun 2025 – Sep 2025

Microsoft Research, Mentors: Christian Bird, Robert DeLine, Carmen Badea

Redmond, WA

- Led a large-scale mixed-methods study with Microsoft developers (N=860) to identify where AI delivers real value in software engineering and how to design it responsibly so developers retain agency, craft, and meaningful work. Built a cognitive task-appraisal framework tied to AI openness and use, identified high-need/low-use targets for tooling investment, and specified 22 AI systems developers want built with the Responsible AI guardrails needed to ship them (ICSE-SEIP 2026; ICSME 2026 Distinguished Paper Award).
- Conducted eight large-scale experiments with 2,000 developers testing contextual transparency patterns and cognitive-forcing interventions; showed that targeted friction improved verification and reduced over-reliance while preserving UX, and produced causal design guidelines for appropriate reliance.
- Co-designed a matched-pair study of individual and team AI usage patterns; identified the Productivity Pressure Paradox (rising expectations without enablement erode benefits) and delivered organizational and product levers to unlock durable gains (ICSE 2026 Distinguished Paper Award).
- Translated findings into product practices with AI for Healthcare, AI Ethics, and Copilot Bing Search, delivering actionable guidance and evaluation rubrics adopted by partner teams.

MOSIP Global Impact Research Fellow (Gates Foundation-funded fellowship)

Mar 2025 – Jun 2025

Modular Open Source Identity Platform (MOSIP), IIIT Bangalore

Corvallis, OR

- Designed an end-to-end multi-agent workflow to automatically debug usability issues in national ID systems; operationalized inclusivity decision rules into AI-driven checks, achieving 87% accuracy on real-world data.
- Built an extensible framework adaptable to diverse personas and inclusivity dimensions (cognitive, socio-economic, gender), providing actionable recommendations to improve equity in large-scale national identity platforms.

Research Scholar, AI4ICPS Lab

May 2022 – Jul 2022

Indian Institute of Technology (IIT) Kharagpur, Advisor: Manoj Sharma

Kharagpur, India

- Led research on trustworthy cyber-physical systems; developed structure-based model defenses against adversarial attacks on autonomous driving robots (CVIP 2022).

Research Student (REU)

Nov 2021 – Apr 2022

Oregon State University, Advisor: Anita Sarma

Remote

- Co-developed the Automated Inclusivity Detector (AID) tool to detect cognitive-bias bugs that impact diverse students in online CS course-ware, and co-designed the user study evaluating it; fixes significantly improved students' control over their learning experience (ICER 2022, ICER 2024).

Associate Software Developer Intern

Sep 2021 – Dec 2021

Nomura Research Institute Financial Technologies India Pvt. Ltd.

Kolkata, India

- Co-developed an automated reconciliation solution enabling extensible, configurable processes for large multi-format databases, improving functionality and efficiency across business operations.

Research Student

Mar 2019 – Jul 2022

St. Thomas' College of Engineering & Technology, Advisors: Amiya Halder, Amit Paul

Kolkata, India

- Developed image processing, soft computing, statistical machine learning, and deep learning algorithms for biomedical and radiological diagnostics, satellite imaging, and robotics; two distinguished paper awards and a Neural Computing and Applications journal article.

Skills

UX Research Methods: Mixed-Methods Research, Field and User Studies, Survey Design and Validation, Semi-Structured Interviews, Contextual Inquiry, Usability Testing, A/B Testing, Experimental and Quasi-Experimental Design, Factorial and Vignette Studies, Longitudinal and Case Studies, Heuristic Evaluation, Cognitive Walkthrough, Inclusive Design, Personas, Participatory Design, Product and Market Research, Grounded Theory, Thematic Analysis, Qualitative Coding

Quantitative and Causal Analysis: Hypothesis Testing, Regression Analysis, Multilevel and Mixed-Effects Models, PLS-SEM, CB-SEM, Exploratory and Confirmatory Factor Analysis, Psychometric Scale Development and Validation, Mediation and Moderation Analysis, Bayesian Statistics, Causal Inference (Randomized Experiments, Quasi-Experimental Methods), Statistical Modeling, Data Mining of Software Repositories and Telemetry

Machine Learning and Generative AI: Supervised and Unsupervised Learning, Feature Engineering, Deep Learning (ANN, CNN, RNN, Transformers, Attention, Adversarial Networks), NLP, Computer Vision, Image Segmentation, LLM Fine-Tuning, Prompt Engineering, LLM Evaluation and Harness Engineering, LLM-as-a-Judge, RAG, Neural Retrieval, Agentic and Multi-Agent Workflows, Agent Orchestration and Coordination, Tool Use/Function Calling, LangGraph, MCP, A2A

Human-Centered AI Design: AI UX Design Patterns, Human-AI Interaction Guidelines, Appropriate Trust and Reliance, Transparency and Explainability, Responsible AI Evaluation, Cognitive Engagement, Work Design and Task Appraisal, Design Systems and Pattern Libraries

Programming, Frameworks, and Tools: Python, R, C/C++, Java, JavaScript, TypeScript, SQL, MATLAB, LaTeX, HTML/CSS, Scikit-Learn, TensorFlow, PyTorch, Keras, OpenCV, Flask, React, Node.js, Git, AWS, Qualtrics, SurveyMonkey, Atlas.ti, Figma, RStudio, JASP, SmartPLS

Selected Publications

33 total publications: 21 conference papers, 5 journal articles, 1 magazine article, 1 book chapter, 5 preprints/under review. Four distinguished paper awards (🏆). Full list on [Google Scholar](#); [pdf] and [data] are hyperlinks.

Under Review

What Makes a Great Co-Worker in an AI-Native Workplace? [pdf | data]

Under Review, 2026

R Choudhuri, M Meijer, SYT Lee, C Lee, C Mannion, P Jahn, A Sarma, C Bird, A Ferng

Microsoft, N=1,556 knowledge workers

You Shall Not Pass! Where and Why Developers Draw the Line on AI Autonomy [pdf | data]

Under Review, 2026

R Choudhuri, C Bird, C Badea, M Gerosa, A Sarma

Microsoft, N=448 developers

Thinking Less, Trusting More: GenAI's Impacts on Students' Cognitive Habits [pdf | data]

Under Review, 2026

R Choudhuri, C Sanchez, M Burnett, A Sarma

5 universities, N=299 students

Journal and Magazine Articles

What Needs Attention? Prioritizing Drivers of Developers' Trust and Adoption of Generative AI [pdf | data]

ACM TOSEM 2026

R Choudhuri, B Trinkenreich, R Pandita, E Kalliamvakou, I Steinmacher, M Gerosa, C Sanchez, A Sarma

Impact Factor: 6.9

The AI-Native Developer: Redefining Work, Identity, and the Future of Craft [pdf]

ACM Queue 2026

R Choudhuri, E Kalliamvakou, B Houck, T Zimmermann

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Adaptive Rough Fuzzy Conditional Spatial Clustering Algorithms for Noisy Brain Tissue Segmentation [pdf]

A Halder, R Choudhuri, A Sarkar

Applied Soft Computing 2026

Impact Factor: 7.8

Measuring SES-Related Traits Relating to Technology Usage: Two Validated Surveys [pdf]

C Chikezie, P Chanpaisaeng, P Agarwal, S Afroz, B Madhwani, R Choudhuri, A Anderson, P Velhal, P Morreale, C Bogart, A Sarma, M Burnett

EMSE 2025

Impact Factor: 4.3

Brain MRI Tumour Classification Using Quantum Classical Convolutional Neural Network Architecture [pdf]

R Choudhuri, A Halder

NCAA 2023

Impact Factor: 4.5

Conference Papers

🏆 To Copilot and Beyond: 22 AI Systems Developers Want Built [pdf | data]

R Choudhuri, C Bird, C Badea, A Sarma. [IEEE Distinguished Paper Award]

ICSME-Industry 2026

Acceptance: 37%

🏆 “Maybe We Need Some More Examples”: Individual and Team Drivers of Developer GenAI Tool Use [pdf | data]

C Miller, R Choudhuri, M Ulloa, S Haniyur, R DeLine, MA Storey, E Murphy-Hill, C Bird, J Butler

ICSE 2026

Acceptance: 21.8%

[ACM SIGSOFT Distinguished Paper Award]

AI Where It Matters: Where, Why, and How Developers Want AI Support in Daily Work [pdf | data | dashboard]

R Choudhuri, C Badea, C Bird, J Butler, R DeLine, B Houck

ICSE-SEIP 2026

Acceptance: 37%

What Guides Our Choices? Modeling Developers’ Trust and Behavioral Intentions Towards GenAI [pdf | data]

R Choudhuri, B Trinkenreich, R Pandita, E Kalliamvakou, I Steinmacher, M Gerosa, C Sanchez, A Sarma

ICSE 2025

Acceptance: 21.2%

Investigating the Impact of Interpersonal Challenges on Feeling Welcome in OSS [pdf]

B Trinkenreich, Z Feng, R Choudhuri, M Gerosa, A Sarma, I Steinmacher

ICSE 2025

Acceptance: 21.2%

Insights from the Frontline: GenAI Utilization Among Software Engineering Students [pdf | data]

R Choudhuri, A Ramakrishnan, A Chatterjee, B Trinkenreich, I Steinmacher, M Gerosa, A Sarma

CSEE&T 2025

Acceptance: 31%

How Far Are We? The Triumphs and Trials of Generative AI in Learning Software Engineering [pdf | data]

R Choudhuri, D Liu, I Steinmacher, M Gerosa, A Sarma

ICSE 2024

Acceptance: 21.2%

Debugging for Inclusivity in Online CS Courseware: Does It Work? [pdf | data]

A Chatterjee, R Choudhuri, M Sarkar, S Chattopadhyay, D Liu, S Hedao, M Burnett, A Sarma

ICER 2024

Acceptance: 20.1%

Structure-Based Learning for Defense Against Adversarial Attacks in Autonomous Driving Agents [pdf]

MK Sharma, R Choudhuri, M Dixit, M Sarkar, B Dittakavi

CVIP 2022

Acceptance: 33%

🏆 Adaptive Rough-Fuzzy Kernelized Clustering Algorithm for Noisy Brain MRI Tissue Segmentation [pdf]

R Choudhuri, A Halder. [IAPR Distinguished Paper Award]

CVIP 2021

Acceptance: 26.1%

🏆 High-Density Salt and Pepper Noise Removal Algorithm Using Statistical Approach [pdf]

A Halder, R Choudhuri. [Best Paper Award]

ICACA 2021

Acceptance: 29.7%

Book Chapter

Automated Brain Tumor Analysis Using Deep Learning Based Framework [pdf]

A Halder, A Sarkar, R Choudhuri. In: Medical Data Analysis and Processing Using Explainable AI

CRC Press 2023

Awards, Grants, and Honors

- 🏆 ACM SIGSOFT Distinguished Paper Award, ICSE 2026
- 🏆 IEEE Distinguished Paper Award, ICSME 2026 (Industry Track)
- 🏆 IAPR Distinguished Paper Award, CVIP 2021
- 🏆 Best Paper Award, ICACA 2021
- MOSIP Global Impact Inclusive Research Fellowship (Gates Foundation-funded), 2025
- NSF Travel Award, ICSE 2024 and ICSE 2025
- OpenAI Researcher Access Program Grant, Sole PI, 2025
- 🏆 Gold Medalist, NPTEL (IIT Ropar); Rank 1 in college, B.Tech Computer Science

Invited Talks and Presentations

What Makes a Great Co-Worker in an AI-Native Workplace? <i>Internship Talk, Tech Futures Group, Microsoft</i>	Aug 2026
AI Where It Matters: Where, Why, and How Developers Want AI Support in Daily Work <i>Internship Talk, SAINTES Group, Microsoft Research</i>	Sep 2025
What Needs Attention? Designing for Appropriate Trust in AI <i>Invited Talk, Appropriate Reliance Group, Microsoft Research</i>	Jul 2025
What Guides Our Choices? Modeling Developers' Trust and Adoption Towards GenAI <i>International Conference on Software Engineering (ICSE) 2025</i>	May 2025
Insights from the Frontline: GenAI Utilization Among Software Engineering Students <i>IEEE Conference on Software Engineering Education and Training (CSEE&T) 2025</i>	Apr 2025
Cognitive Factors Affecting Trust and Adoption Towards AI <i>Invited Talk, Colorado State University</i>	Nov 2024
How Far Are We? The Triumphs and Trials of Generative AI in SE <i>International Conference on Software Engineering (ICSE) 2024</i>	Apr 2024

Selected Media and Industry Coverage

Media Coverage

AI might be reshaping how college students learn, at the expense of critical thinking KGW TV News (Link)	May 2026
OSU study raises concerns about AI's impact on student thinking skills KATU ABC2 TV News (Link)	May 2026
OSU study warns of declining cognitive ability as AI use grows in STEM fields KOIN 6 News (Link); Yahoo News (Link)	May 2026
The staggering 'cognitive debt' of generative AI in education Jefferson Public Radio, The Jefferson Exchange (Link)	May 2026
America's STEM students are falling into an 'AI dependence spiral' SAN News (Link)	May 2026
Tech-savvy STEM students prone to letting AI do their thinking for them The News Guard (Link)	May 2026
AI literacy to prevent scams in the AI era KGW TV News (Link)	May 2026

Industry Coverage

Where do developers draw the line on AI autonomy? Research-Driven Engineering Leadership (RDEL #158); features "You Shall Not Pass" (Link)	Aug 2026
Five studies that are changing how I think about AI in software engineering Engineering Enablement (DX), Brian Houck; features "AI Where It Matters" and "To Copilot and Beyond" (Link)	Jul 2026
The AI-native developer Engineering Enablement (DX), Brian Houck; features "The AI-Native Developer", ACM Queue (Link)	May 2026
How does deepening AI fluency change what it means to be a software developer? Research-Driven Engineering Leadership (RDEL #143); features "The AI-Native Developer" (Link)	May 2026
Where do developers actually want AI to support their work? Research-Driven Engineering Leadership (RDEL #114); features "AI Where It Matters" (Link)	Oct 2025
What distinguishes developers who successfully adopt AI tools from those who don't? Research-Driven Engineering Leadership (RDEL #108); features "Maybe We Need Some More Examples" (Link)	Sep 2025
What influences developers' trust in adopting AI-assisted coding tools? Research-Driven Engineering Leadership (RDEL #84); features "What Guides Our Choices?" (Link)	Mar 2025

Professional Service and Mentoring

Mentoring, graduate students: [Ambareesh Ramakrishnan](#), [Sadia Afroz](#), [Olivia Schutz](#), [Fatima A. Moussaoui](#), [Yunhan Qiao](#), [Chifang Chou](#)

Mentoring, undergraduate students: [Dylan Liu](#), [Mrinmoy Sarkar](#), [Pierce Fleming](#), [Arinjay Bhowmick](#), [Mayank Dixit](#)

Reviewer: IEEE Transactions on Software Engineering (TSE); Information and Software Technology (IST); Journal of Systems and Software (JSS); ACM CHI 2024; ACM Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP) 2022, 2023, 2024; International Conference on Computer Vision and Image Processing (CVIP) 2022, 2023

Sub-reviewer: ICPC 2025; ICSE 2024; MSR 2024; ESEC/FSE 2023; TSE; EMSE